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Internship completion confirmation - RTL Code Generator for VLSI

1 message

Roshan Dsouza(UST,IN) <Roshan.Dsouza@ust.com> Tue, May 26, 2026 at 11:12 PM
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Dear Sir/ Madam,

This is to certify that the following students of **Vishwakarma Institutes** have successfully completed the industry project titled **"RTL Code Generator For VLSI"** under the valuable guidance and mentorship of:

- **Mentors from UST: Mr.** Roshan Dsouza
- **Mentors from Vishwakarma institute:** Dr. Anuradha Yenkikar, Dr. Abhay Chopde

Student Details

1. **Sarvesh Warule (22420229)**
2. **Nitin Panzade (22420267)**
3. **Akash Patel (22310745)**
4. **Shreyas Phansalkar(12311546)**
5. **Sampada Sanap (12310288)**
6. **Pushpa (12310278)**

The project was carried out during the period from **05/01/2026 to 05/04/2026**. The work was completed successfully in accordance with the proposed objectives, project scope, and timeline.

The project involved the effective implementation of an AI-driven automated RTL code generation system for VLSI design. The system integrates advanced RTL generation, intelligent testbench generation via Groq API, dataset matching with 439 pre-tested samples, and comprehensive Icarus Verilog simulation workflows.

Key components including the testbench generator, dataset matcher with multi-factor similarity scoring, RTL validator, and simulation pipeline optimizer demonstrate sincere efforts toward innovation in hardware design automation. The 4-tier fallback architecture—combining AI generation, dataset matching, reference designs, and generic templates—achieves 90-100% success rates with 75-90% code quality.

We appreciate the efforts of the student team and the consistent mentorship provided by both guides. We commend all contributors for their commitment to academic excellence and innovation in advancing automated hardware verification and design automation.

Regards,
Roshan